

LABORATÓRIO 1

```
jps
start-dfs.sh
hadoop fs -mkdir -p /user/bigdata/raw/customers
hadoop fs -mkdir -p /user/bigdata/raw/transactions
hadoop fs -mkdir -p /user/bigdata/raw/fraud_labels
hadoop fs -mkdir -p /user/bigdata/bronze
hadoop fs -mkdir -p /user/bigdata/silver
hadoop fs -mkdir -p /user/bigdata/gold

hadoop fs -ls -R /user/bigdata

/user/bigdata/bronze
/user/bigdata/gold
/user/bigdata/raw
/user/bigdata/raw/customers/customers_synthetic.csv      842451 bytes
/user/bigdata/raw/fraud_labels/fraud_labels.csv          218900 bytes
/user/bigdata/raw/transactions/transactions_synthetic.csv 8139648 bytes
/user/bigdata/silver
```

Foi criada e conferida a estrutura de camadas no HDFS, com as pastas raw, bronze, silver e gold. Os três arquivos de dados foram armazenados nas respectivas pastas da camada raw.

Verificação do arquivo no HDFS

```
hdfs fsck /user/bigdata/raw/transactions/transactions_synthetic.csv -files -blocks -locations
```

```
Status: HEALTHY
Arquivo: transactions_synthetic.csv
Tamanho: 8139648 bytes
Blocos: 1
Replication: 1
DataNodes: 1
Missing blocks: 0
Corrupt blocks: 0
```

Contagem de registros

```
hadoop fs -cat /user/bigdata/raw/customers/*.csv | wc -l
hadoop fs -cat /user/bigdata/raw/transactions/*.csv | wc -l
```

```
9994
100001
```

As contagens incluem o cabeçalho dos arquivos CSV. Portanto, correspondem a 9.993 clientes e 100.000 transações de dados.